

JOB SHEET

DECISION SUPPORT SYSTEM

MEETING : 6
MATERIAL : ELECTRE IV
PURPOSE : Students understand the problems to be solved with the Electre IV method

1. Discussion

Electre is one of the multicriteria decision-making methods based on the concept of outranking by using a paired comparison of alternatives based on each appropriate criterion.

The Electre method is used in conditions where the less than the criteria alternatives will be eliminated, and suitable alternatives can be generated, in other words Electre is used for cases with many alternatives. However, few criteria are involved.

An alternative is said to dominate another alternative if one or more of its criteria exceed (compared to the criteria of the other alternative) and are equal to the remaining criteria '(Kusumadewi, 2006)'.

The steps taken in solving problems using the Electre method are as follows:

1. Normalization of the decision matrix

In this procedure, each attribute is converted into a comparable value. Each r_{ij} normalization can be done with equation (1):

$$r_{ij} = \frac{x_{ij}}{\sqrt{\sum_{i=1}^m x_{ij}^2}}, \text{ Untuk } i = 1, 2, 3, \dots, m \text{ dan } j = 1, 2, 3, \dots, n$$

Keterangan :

r_{ij} = normalisasi pengukuran pilihan dari alternatif dan kriteria.
 m = Alternatif.
 n = Kriteria.

(1)

So that the normalized R matrix is obtained:

$$R = \begin{bmatrix} r_{11} & r_{12} & \dots & r_{1n} \\ r_{21} & r_{22} & \dots & r_{2n} \\ r_{m1} & r_{m2} & \dots & r_{mn} \end{bmatrix}$$

(2)

R is a matrix that has been normalized or called a normalized decision matrix. Where m states the alternative, n states the criteria and r_{ij} is the normalization of the choice measurement of the first alternative in relation to the second criterion.

2. Weighting on normalized matrix

Once normalized, each column of the matrix R is multiplied by the weights (w_j) determined by the decision-maker. Thus, the weighted normalized matrix is $V = RW$ is the written following equation.

$$V = \begin{bmatrix} v_{11} & v_{12} & \dots & v_{1n} \\ v_{21} & v_{22} & \dots & v_{2n} \\ \dots & \dots & \dots & \dots \\ v_{m1} & v_{m2} & \dots & v_{mn} \end{bmatrix} = RW = \begin{bmatrix} w_1 r_{11} & w_2 r_{12} & \dots & w_n r_{1n} \\ w_1 r_{21} & w_2 r_{22} & \dots & w_n r_{2n} \\ \dots & \dots & \dots & \dots \\ w_1 r_{m1} & w_2 r_{m2} & \dots & w_n r_{mn} \end{bmatrix}$$

Where W is:

$$W = \begin{bmatrix} w_1 & 0 & 0 & \dots & 0 \\ 0 & w_2 & 0 & \dots & 0 \\ \dots & \dots & \dots & \dots & \dots \\ 0 & 0 & 0 & \dots & w_n \end{bmatrix}, \text{ dan } \sum_{i=1}^n w_i = 1 \quad (3)$$

3. Determining concordance and discordance index

For each pair of the alternatives k and l ($k, l = 1, 2, 3, \dots, \mu$ and $k \neq l$) the set of criteria J is divided into two subsets, namely concordance and discordance. If a criterion in an alternative including concordance is:

$$C_{kl} = \{j, v_{kj} \geq v_{lj}\}, \text{ untuk } j = 1, 2, 3, \dots, n \quad (4)$$

Conversely, the opposite of this subset is discordance, i.e. when:

$$D_{kl} = \{j, v_{kj} < v_{lj}\}, \text{ untuk } j = 1, 2, 3, \dots, n \quad (5)$$

Keterangan :

C_{kl} = himpunan *concordance*.

D_{kl} = himpunan *discordance*.

v_{kj} = indeks dari matriks V .

v_{lj} = indeks dari matriks V .

4. Calculate concordance and discordance matrix

a. Concordance

To determine the value of the elements in the concordance matrix is to sum the weights included in the concordance subset:

$$c_{kl} = \sum_{j \in C} w_j \quad (6)$$

So that the resulting concordance matrix is:

$$C = \begin{bmatrix} - & c_{12} & c_{13} & \cdots & c_{1n} \\ c_{21} & \cdots & c_{23} & \cdots & c_{2n} \\ \cdots & \cdots & \cdots & \cdots & \cdots \\ c_{m1} & c_{m2} & c_{m3} & \cdots & - \end{bmatrix}$$

b. Discordance

To determine the value of the elements in the discordance matrix is to divide the maximum difference in the value of the criteria included in the discordance subset by the maximum difference in the value of all existing criteria, mathematically speaking:

$$d_{kl} = \frac{\{\max(v_{mn} - v_{mn-l_n})\}_{m,n \in d_{kly}}}{\{\max(v_{mn} - v_{ln})\}_{m,n = 1,2,3,\dots}} \quad (7)$$

Next, a discordance matrix is obtained:

$$D = \begin{bmatrix} - & d_{12} & d_{13} & \cdots & d_{1n} \\ d_{21} & \cdots & d_{23} & \cdots & d_{2n} \\ \cdots & \cdots & \cdots & \cdots & \cdots \\ d_{m1} & d_{m2} & d_{m3} & \cdots & - \end{bmatrix}$$

5. Determining the dominant matrix of concordance and discordance

a. Concordance

A concordance dominant matrix can be constructed with the help of threshold values, i.e. by comparing each value of the concordance matrix element with the threshold value.

$$c_{kl} \geq \underline{c}$$

Dengan nilai *threshold* ©, adalah :

$$\underline{c} = \frac{\sum_{k=1}^n \sum_{l=1}^n c_{kl}}{m * (m - 1)} \quad (8)$$

And each element of the matrix F as the dominant matrix of concordance is defined as follows:

$$f_{kl} = 1, \text{ jika } c_{kl} \geq \underline{c} \text{ dan } f_{kl} = 0, \text{ jika } c_{kl} < \underline{c} \quad (9)$$

b. Discordance

To build a discordance dominant matrix, we also use the help of threshold values, namely:

$$\underline{d} = \frac{\sum_{k=1}^n \sum_{l=1}^n d_{kl}}{m * (m - 1)} \quad (10)$$

And the value of each element for matrix G as a dominant matrix of discordance is determined as follows:

$$g_{kl} = 1 \text{ , jika } d_{kl} \geq \underline{d} \text{ dan } g_{kl} = 0 \text{ , jika } d_{kl} < \underline{d} \quad (11)$$

6. Define the aggregate dominance matrix

The next step is to determine the aggregate dominance matrix as the matrix E, each element of which is a multiplication between the element of the matrix F and the element of the matrix G, as follows:

$$e_{kl} = f_{kl} * g_{kl} \quad (12)$$

7. Elimination of less favourable alternatives

The E matrix gives the order of choice of each alternative, i.e. if $e = 1$ kl then the alternative Ak is a better choice than Ar so that the row in the E matrix that has the least amount of $e = 1$ kl can be eliminated. Thus the best alternative is the one that dominates the other alternatives.

2. Case Examples

An example of the application of Electre in the scholarship acceptance decision support system. With the criteria of GPA, parental income per year, number of dependents and parental work. The steps are as follows.

1. Weighting criteria

Criterion weighting is carried out to determine the absolute value of the weight of each criterion so that it can be easily assimilated into the ELECTRE method used.

This weighting is done by experts or interested parties.

Tabel 1. Bobot IPK	
Nilai IPK (K1)	Bobot
>3,5	5
3,1 – 3,5	4
2,6 – 3,0	3
2,0 – 2,5	2
<2,0	1

Tabel 3. Bobot Jumlah Tanggungan	
Jumlah Tanggungan (K3)	Bobot
<2	1
2-3	2
4-5	3
6-7	4
>7	5

Tabel 2. Bobot Penghasilan per Tahun	
Penghasilan per Tahun (K2)	Bobot
>36.000.000	1
24.000.001 – 36.000.000	2
18.000.001 – 24.000.000	3
12.000.000 – 18.000.000	4
<12.000.000	5

Tabel 4. Bobot Pekerjaan Orang Tua	
Pekerjaan Orang Tua (K4)	Bobot
1-2	1
3-3	2
4-4	3
5-5	4
6	5

2. Determination of a sample of prospective scholarship recipients

Then an alternative data sample was collected, which was scored based on the above provisions.

Tabel 5. Calon Penerima Beasiswa				
Calon Penerima Beasiswa	Nilai Kriteria			
	K1	K2	K3	K4
Mahadi	5	3	3	3
Saimon	4	4	3	4
Wahdi	4	4	3	3
Aditya	2	4	3	3

So the alternative data table is:

	c1	c2	C3	C4
a1	5	3	3	3
a2	4	4	3	4
a3	4	4	3	3
A4	2	4	3	3

3. Determination of weight

Next, the weight (W) of each criterion is determined.

c1	c2	C3	C4
5	2	4	2

4. Normalization of alternative matrices

Normalization is carried out according to equation (1).

$$R_{11} = \frac{5}{\sqrt{5^2 + 4^2 + 4^2 + 2^2}} = 0.640$$

$$R_{21} = \frac{4}{\sqrt{5^2 + 4^2 + 4^2 + 2^2}} = 0.512$$

$$R_{31} = \frac{4}{\sqrt{5^2 + 4^2 + 4^2 + 2^2}} = 0.512$$

$$R_{41} = \frac{2}{\sqrt{5^2 + 4^2 + 4^2 + 2^2}} = 0.256$$

And so on to R44, so that the matrix of the normalization result of R is obtained:

R normalization matrix				
R	c1	c2	C3	C4
a1	0.640	0.397	0.500	0.457
a2	0.512	0.530	0.500	0.610
a3	0.512	0.530	0.500	0.457
A4	0.256	0.530	0.500	0.457

5. Weighting on normalized matrix

Multiply each cell in the normalized matrix with the weight of the criteria according to equation (3), so that matrix V is obtained:

Normalization matrix V

weight:	5	2	4	2
V	c1	c2	C3	C4
a1	3.201	0.795	2.000	0.915
a2	2.561	1.060	2.000	1.220
a3	2.561	1.060	2.000	0.915
A4	1.280	1.060	2.000	0.915

6. Matrix Concordance

- a. To determine the concordance of the matrix, first determine the set of concordances with equation (4).

$C_{12} = V_{11} \geq V_{21} \rightarrow 3.201 \geq 2.561 \rightarrow \text{Criterion 1} \rightarrow V_{12}$
 $\text{Concordance} \geq V_{22} \rightarrow 0.768 \geq 1.024 \rightarrow \text{Criterion 2} \rightarrow -$
 $V_{13} \geq V_{23} \rightarrow 1.536 \geq 1.536 \rightarrow \text{Criterion 3} \rightarrow V_{14}$
 $\text{Concordance} \geq V_{24} \rightarrow 0.768 \geq 1.024 \rightarrow \text{Criterion 4} \rightarrow -$
 So $C_{12} = \{1, 3\}$

And so on to C43, until the concordance assembly is obtained:

	Index Criteria
c12	1,3
C13	1,3,4
c14	1,3,4
c21	2,3,4
C23	1,2,3,4
c24	1,2,3,4
C31	2,3,4
c32	1,2,3
C34	1,2,3,4
C41	2,3,4
C42	2,3
c43	2,3,4

- b. Next, add up the value of the alternative criteria according to the criterion index in the concordance set, according to equation (6).

$C_{12} = W_1 + W_3 \rightarrow 5 + 4 = 9$
 $C_{13} = W_1 + W_3 + W_4 \rightarrow 5 + 4 + 2 = 11$
 $C_{14} = W_1 + W_3 + W_4 \rightarrow 5 + 4 + 2 = 11$

And so on to C43, then arranged into the concordance matrix C:

Concordance Matrix C				
C	a1	a2	a3	A4
a1	0	9	11	11
a2	8	0	13	13
a3	8	11	0	13
A4	8	6	8	0

- c. Find the threshold concordance (c). That is, sum all the values of the concordance matrix C, then divide by $m*(m-1)$ where m is the alternative number, based on equation (8).

$$c = \frac{9 + 11 + 11 + 8 + 13 + 13 + 8 + 11 + 13 + 8 + 6}{4 * (4 - 1)} = 9.92$$

- d. The next step is to find the dominant matrix concordance F, which is to compare each value of matrix C with the threshold. If it is greater than the threshold, then give it a value of 1.

Concordance dominant matrix F				
F	a1	a2	a3	A4
a1	0	0	1	1
a2	0	0	1	1
a3	0	1	0	1
A4	0	0	0	0

7. Matrix Discordance

- a. To determine the discordance of the matrix, first determine the set of discordance with equation (5).

$$\begin{aligned}
 D12 &= V11 \geq V21 \rightarrow 3,201 < 2,561 \rightarrow \text{Criterion 1} \rightarrow - \\
 &V12 \geq V22 \rightarrow 0.795 < 1.060 \rightarrow \text{Criterion 2} \rightarrow V13 \\
 &\text{discordance} \geq V23 \rightarrow 2,000 < 2,000 \rightarrow \text{Criterion 3} \rightarrow - \\
 &V14 \geq V24 \rightarrow 0.915 < 1.220 \rightarrow \text{Criterion 4} \\
 &\rightarrow \text{discordance up to } D12 = \{2, 4\}
 \end{aligned}$$

And so on to D43, until the concordance assembly is obtained:

	Index Criteria
d12	2,4
d13	2
D14	2
D21	1
D23	-
D24	-
D31	1
D32	4
D34	-
D41	1
D42	1,4
D43	1

- b. Next, look for the absolute max value of the criterion value according to the concordance set divided by the absolute max of the criterion value, in each alternative. According to equation (7).

$$D12 = \frac{\max\{|0.795 - 1.060|; |0.915 - 1.220|\}}{\max\{|3.201 - 2.561|; |0.795 - 1.060|; |2,000 - 2,000|; |0.915 - 1.220|\}}$$

$$\begin{aligned}
&= \frac{\max\{0.265; 0.305\}}{\max\{0.64; 0.265; 0; 0.3\}} \\
&= \frac{0.305}{0.64} \\
&= 0.476
\end{aligned}$$

$$\begin{aligned}
D_{13} &= \frac{\max\{|0.795 - 1.060|\}}{\max\{|3.201 - 2.561|; |0.795 - 1.060|; |2,000 - 2,000|; |0.915 - 1.220|\}} \\
&= \frac{\max\{0.265; \}}{\max\{0.64; 0.265; 0; 0.3\}} \\
&= \frac{0.265}{0.64} \\
&= 0.414
\end{aligned}$$

And so on to D₄₃. It is then arranged into a discordance matrix D

D	a1	a2	a3	A4
a1	0	0.476421	0.413797	0.137932
a2	1	0	0	0
a3	1	1	0	0
A4	1	1	1	0

- c. Find the discordance threshold ($\underline{d}$), which is to sum all the values of the discordance matrix D, then divide by $m*(m-1)$ where m is the alternate number, based on equation (10).

$$c = \frac{0.476 + 0.414 + 0.138 + 1 + 1 + 1 + 1 + 1 + 1}{4 * (4 - 1)} = 0.585$$

- e. The next step is to find the dominant matrix of discordance F, which is to compare each value of matrix D with the threshold. If Larger from the threshold, then give a value of 1.

Dominant matrix discordance G				
G	a1	a2	a3	A4
a1	0	0	0	0
a2	1	0	0	0
a3	1	1	0	0
A4	1	1	1	0

8. Aggregation matrix

To find the aggregation matrix E, is the multiplication of the matrix F by G according to the equation (12).

Matriks dominan concordance F				
F	a1	a2	a3	a4
a1	0	0	1	1
a2	0	0	1	1
a3	0	1	0	1
a4	0	0	0	0

X

Matriks dominan discordance G				
G	a1	a2	a3	a4
a1	0	0	0	0
a2	1	0	0	0
a3	1	1	0	0
a4	1	1	1	0

Matriks agregasi E				
E	a1	a2	a3	a4
a1	0	0	0	0
a2	0	0	0	0
a3	0	1	0	0
a4	0	0	0	0

9. Conclusion

- Look for the line with the most 1s.
- In this case, a3 is the choice. So alternative 3, namely Wahdi, is the one who is most entitled to receive a scholarship.

ASSIGNMENT

Create case resolution using the ELECTRE IV Method with free cases with case conditions:

- There are at least 4 criteria
- There are at least 5 alternative options

Assignment conditions:

- It is done in groups of 2-3 people, the group is different from the previous jobsheet.
- The tasks collected are (1) pdf format files and (2) calculations in excel files.
- There are many possible numbers that can be used so the numbers should not be the same. Do it yourself in groups.